

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.IT Semester IV)
DATA VISUALIZATION WITH POWERBI AND TABLEAU

Practical – IX

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Aim : Cleaning and Structuring Data for Analysis

A) Identify and handle missing or inconsistent data:

Step 1: Load the Dataset

- . Open Tableau Desktop and load the given dataset.
- . Go to Sheet 1.

Step 2: Clean Physics and Attendance

- . Right-click in the Data Pane - Create Calculated Field.

- . Create Physics_Clean.

Formula: `FLOAT (REPLACE (STR( [Physics] ), '%', ''))`

- . Repeat the same steps and create Attendance_Clean.

Formula : `FLOAT (TRIM (REPLACE (STR ( [Attendance] ), '%', '')))`

The screenshot displays the Tableau Desktop interface. At the top, a text box contains the name 'Physics_clean'. Below it, the formula `FLOAT (REPLACE (STR( [Physics] ), '%', ''))` is entered. On the left, a context menu is open with options: 'Create Calculated Field...', 'Create Parameter...', 'Create Folder (use group by folder)', 'Group by Folder', 'Group by Data Source Table', and 'Sort by Name'. The 'Create Calculated Field...' option is selected. In the center, the 'Tables' pane lists data sources: 'Attendance', 'Dept Info', 'Full Name', 'Student ID', 'Measure Names', 'Chemistry', 'Math', 'Physics', 'Physics_clean' (highlighted with a green bar), 'Programming', 'Sem1_Scores.csv (Count)', and 'Measure Values'. On the right, another 'Tables' pane lists: 'Attendance', 'Dept Info', 'Full Name', 'Student ID', 'Measure Names', 'Attendance_clean' (highlighted with a green bar), 'Chemistry', 'Math', 'Physics', 'Physics_clean', 'Programming', 'Sem1_Scores.csv (Count)', and 'Measure Values'.

Step 3: Handle Missing Values

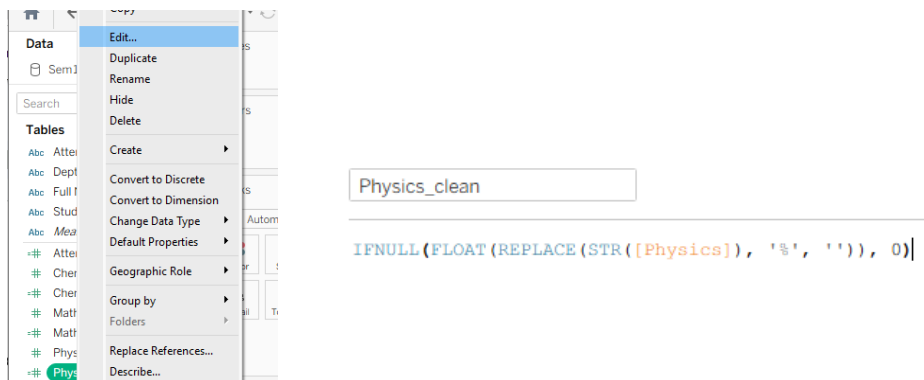
- . Create Math_Clean using `IFNULL([Math], 0)`.
- . Create Chemistry_Clean using `IFNULL([Chemistry], 0)`.



Step 4: Handle Missing Physics Values

. Right-click Physics_Clean - Edit and use:

Formula: IFNULL(FLOAT (REPLACE (STR([Physics]), '%', '')), 0)



B) Convert Wide Data into Tall Format:

Step 1: Open Data Source . Open the Data Source tab.

Step 2: Pivot Columns:

. Hold Ctrl and select Math, Physics, Chemistry, Programming.

. Right-click the selected columns - Pivot.

	Field Name	Physical Tab
	Dept Info	Sem1_Score
	Attendance	Sem1_Score
	Pivot Field Names	Pivot
	Pivot Field Values	Pivot
	Physics_clean	Calculation
	Attendance_clean	Calculation
	Math_clean	Calculation

Step 3: Rename Pivoted Fields

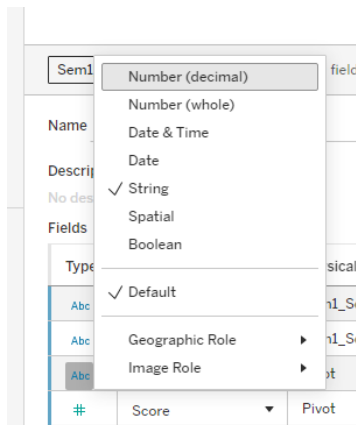
. Rename Pivot Field Names to Subject.

. Rename Pivot Field Values to Score.

Abc	Attendance	S
Abc	Subject	P
#	Score	P
#	Physics_clean	C
#	Attendance_clean	C

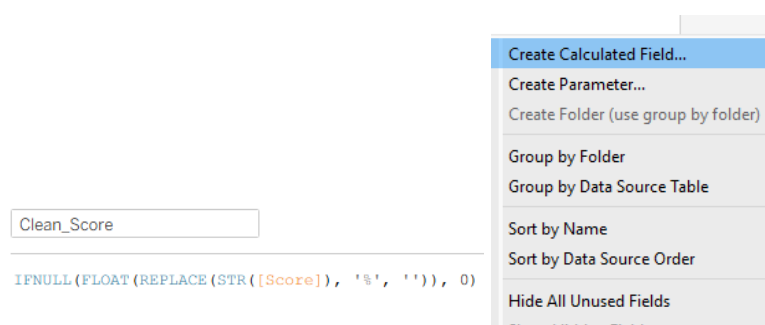
Step 4: Format Score:. Click the Abc icon above Score.

. Select Number (decimal).



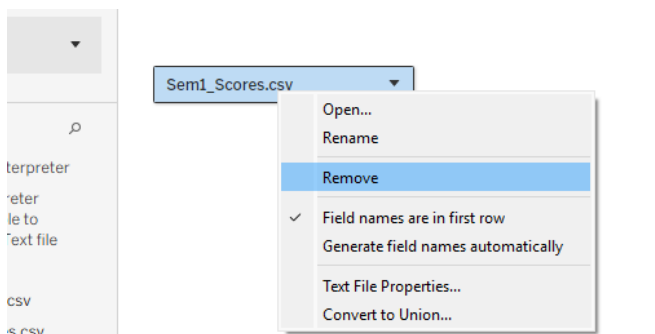
Alternative: Create Clean_Score

Formula: IFNULL (FLOAT (REPLACE (STR([Score]), '%', '')), 0)



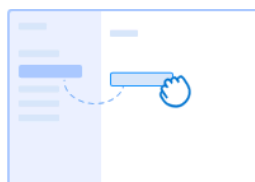
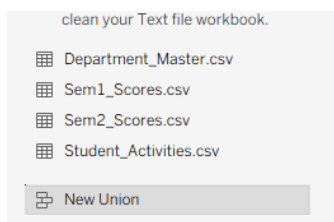
C) Perform Union of Multiple Files:

Step 1: Open the Data: Clear the existing canvas



Step 2: Create a Multi-File Union

- . Drag New Union to the canvas.
- . Add Sem1_Scores.csv and Sem2_Scores.csv.



Drag tables here to create a data m

Connection: Sem1_Scores



Step 3: Verify the Merged Data Grid

- . Check the data grid at the bottom.
- . Confirm that the total is 20 rows.
- . Check the Table Name column for the two files.

Tables in union: 2



Union	14 fields 20 rows	20	rows
Name Union	Abc Union	Abc Union	Abc Union

3. Scroll to the far right of the grid to verify the generated system column: Table Name: Displays "Sem1_Scores.csv" for the first 10 rows and

"Sem2_Scores.csv" for the remaining 10 rows.

Attendance	Table Name	Physics_clean	Attendance
%	Sem1_Scores.csv	0.900000	
%	Sem1_Scores.csv	0.750000	
%	Sem1_Scores.csv	0.820000	
%	Sem1_Scores.csv	0.880000	
%	Sem1_Scores.csv	0.000000	
%	Sem1_Scores.csv	0.910000	
%	Sem1_Scores.csv	0.700000	

Step 4: Create a Clean Semester Identifier Dimension

1. Click on Sheet 1 at the bottom-left.
2. Create Calculated Field.Set the Field Name to: Semester
4. Enter the following formula:

Create Calculated Field...
Create Parameter...
Create Folder (use group by folder)
Group by Folder
Group by Data Source Table

Semester

```

IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END

```

D) Execute cross-database joins:

Step 1: Stay on the Top Canvas. Leave your Union table box as it is on the canvas.

Step 2: Add and Link Department_Master.csv

1. In the left sidebar under Files / Connections, find Department_Master.csv. 2. Drag Department_Master.csv onto the canvas and drop it to the right of your Union box.
3. A flexible line (relationship / join noodle) will connect the two boxes.



4. If the relationship dialog opens (or click the relationship line):
 - o Left Table Field (Union): Select Dept_Info
 - o Right Table Field (Department_Master): Select Dept_Code
 - o Operator: =

Union — Departme...

Name Department_Master.csv

How do relationships differ from joins? [Learn more](#)

Union

Operator

Department_Maste...

Abc Dept_Info

=

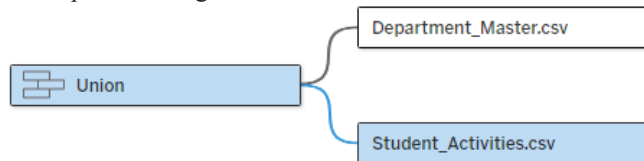
Abc Dept Code

Union

Department_Master.csv

Step 3: Add and Link Student_Activities.csv

1. From the left sidebar, drag Student_Activities.csv onto the canvas.
2. Drop it to the right of the Union box.



3. In the relationship dialog:
 - o Left Table Field (Union): Select Student_ID
 - o Right Table Field (Student_Activities): Select Student_ID
 - o Operator: =

Union — Student_A...

Name Student_Activities.csv

How do relationships differ from joins? [Learn more](#)

Union

Operator

Student_Activities....

Abc Student_ID

=

Abc Student ID (Stu...

Step 4: Verify in Sheet 1

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane on the far left, you will now see all three tables organized with their respective fields:
 - o Union: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name
 - o Department_Master: Dept_Code, Department_Name, HOD_Name, Dept_Target_Average, Total_Dept_Capacity
 - o Student_Activities: Club_Membership, Projects_Completed, Certifications

Tables		
Department_Master.csv Abc Department Name Abc Dept Code Abc HOD Name # Dept Target Average # Total Dept Capacity # Department_Master...	Union Abc Attendance Abc Dept_Info Abc Full_Name =Abc Semester Abc Student_ID Abc Table Name	Attendance_clean
		Chemistry
		Chemistry_Clean
		Math
		Math_clean
		Physics
		Physics_clean
Programming		
Union (Count)		
Measure Names		
Clean_Score		
Measure Values		

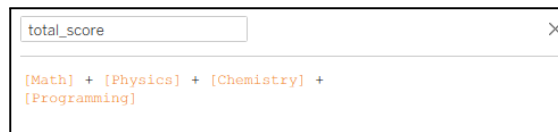
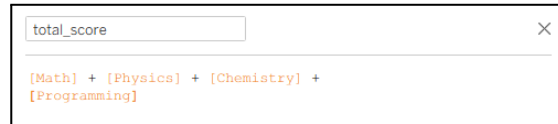
E.) Restructure Data for Better Visualization:

Step 1: Calculate Total Marks

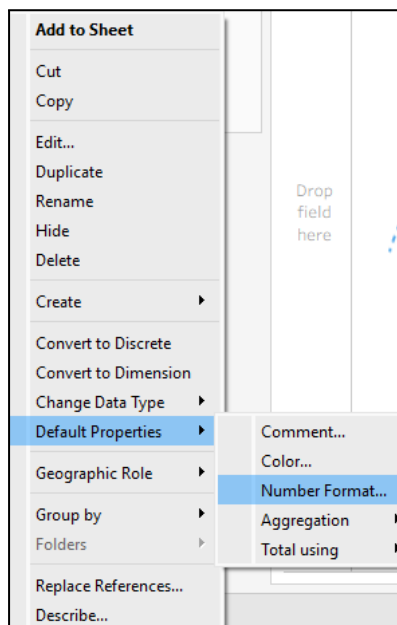
1. Open Tableau.
2. Go to Sheet 1.
3. In the Data Pane, right-click on an empty area.
4. Select Create Calculated Field.
5. Enter field name:
Total_Score
6. Enter formula:
[Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]
7. Click OK.

Step 2: Calculate Overall Percentage

1. Right-click in the Data Pane.
2. Select Create Calculated Field.
3. Field name:
Overall_Percentage
4. Formula:
[Total_Score] / 400.0
5. Click OK.
6. Right-click Overall_Percentage.
7. Select Default Properties → Number
8. Select Percentage.
9. Set decimal places to 1.
10. Click OK.



Format.



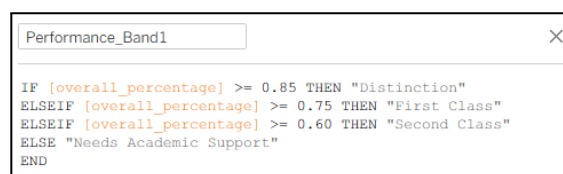
Step 3: Create Performance Band

1. Right-click in Data Pane.
2. Select Create Calculated Field.
3. Field name: Performance_Band

Enter:

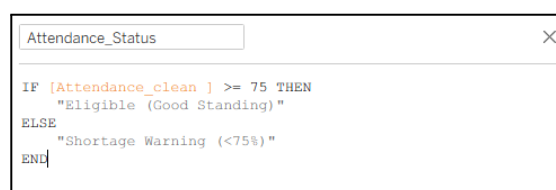
```
IF [Overall_Percentage] >= 0.85 THEN "Distinction"
ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support"
```

4. END
5. Click OK.



Step 4: Create Attendance Status

1. Right-click Data Pane.



2. Select Create Calculated Field.
3. Field name:
Attendance_Status

Formula: IF [Attendance_clean] >= 75 THEN
"Eligible (Good Standing)"

ELSE

"Shortage Warning (<75%)"

END

4. END
5. Click OK.

Step 5: Build Student Performance Scorecard

Now create the visualization:

1. Drag Student_ID → Rows.
2. Drag Full_Name → Rows.
3. Drag Semester → Columns.
4. Drag Overall_Percentage → Text in Marks card.
5. Drag Performance_Band → Color.
6. Drag Attendance_Status → Tooltip or Detail.

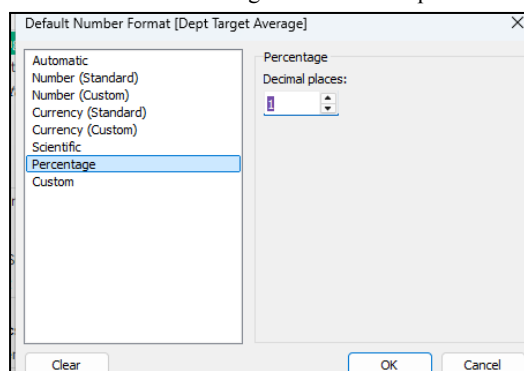
Output:-

Student ID	Full Name	Semester 2
S101	Aarav Sharma	67.2%
S102	priya VERMA	
S103	Rohan Gupta	
S104	Sneha Patel	69.7%
S105	aditya Rao	
S106	Ananya Singh	66.5%
S107	Rahul Nair	
S108	Kavya Reddy	71.5%
S109	Vikram Joshi	49.4%
S110	Pooja Iyer	63.0%

F. Handle Different Levels of Detail Using LOD

Step 1: Department Average

1. Go to Sheet 1 or create Sheet 2.
2. Right-click in Data Pane.
3. Select Create Calculated Field.
4. Field name:
Dept_Average_Percentage_LOD
5. Formula:
{ FIXED [Dept_Info] : AVG([Overall_Percentage]) }
6. Click OK.
7. Right-click the new field.
8. Select Default Properties → Number Format.
9. Select Percentage and 1 decimal place.



Step 2: Student Career Average

1. Right-click Data Pane.
2. Select Create Calculated Field.
3. Field name:
Student_Overall_Career_Average
4. Formula:
{ FIXED [Student_ID] : AVG([Overall_Percentage]) }
5. Click OK.
6. Set the number format to Percentage with 1 decimal place.



Step 3: Calculate Variance

1. Create another Calculated Field.
2. Field name:
Variance_From_Dept_Average
3. Formula:
[Overall_Percentage] -
[Dept_Average_Percentage_LOD]
4. Click OK.
5. Set it to Percentage with 1 decimal place.



Step 4: Create Benchmark Status

1. Create another Calculated Field.
2. Field name: Benchmark_Status

Formula:

IF [Variance_From_Dept_Average] >= 0 THEN
"Above Department Benchmark"
ELSE "Below Department Benchmark"

3. END
4. Click OK.



Step 5: Build LOD Visualization

1. Drag Dept_Info → Rows.
2. Drag Student_ID → Rows.
3. Drag Full_Name → Rows.
4. Drag Semester → Columns.
5. Drag Overall_Percentage → Text.
6. Drag Dept_Average_Percentage_LOD → Tooltip.
7. Drag Benchmark_Status → Color.

Final result

Pages

Columns

Semester

Rows

Dept Info

Student ID

Full Name

Filters

Build LOD Visualization

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Benchmark St.

SUM(Dept Tar.

SUM(overall_p

Dept Info	Student ID	Full Name	Semester	Semeste r 2
CSE-01	Null			
	S101	Aarav Sharma	67.2%	
	S103	Rohan Gupta		
	S106	Ananya Singh	66.5%	
ECE-02	S109	Vikram Joshi	49.4%	
	S102	priya VERMA		
	S107	Rahul Nair		
	S110	Pooja Iyer	63.0%	
IT-03	S104	Sneha Patel	69.7%	
	S108	Kavya Reddy	71.5%	
MECH-04	S105	aditya Rao		

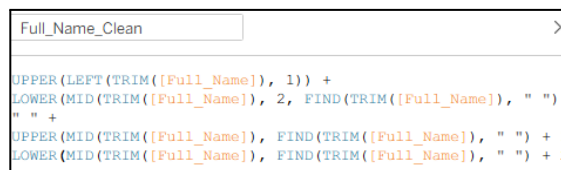
G: Apply Advanced Data Cleaning Techniques

STEP 1 — Full_Name_Clean

1. Sheet 1 par raho.
2. Left side Data Pane mein right-click karo.
3. Create Calculated Field select karo.
4. Field Name: Full_Name_Clean
5. Formula:

UPPER(LEFT(TRIM([Full_Name]), 1)) +

LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") - 1)) +



" " +

UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " " + 1, 1)) +
LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " " + 2, LEN(TRIM([Full_Name])))))

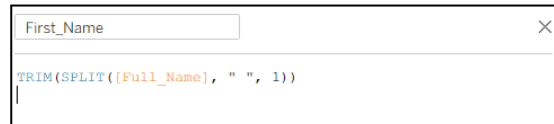
6. OK click karo.

STEP 2 — First_Name

1. Data Pane → Right Click
2. Create Calculated Field
3. Name:
First_Name
4. Formula:

TRIM(SPLIT([Full_Name], " ", 1))

5. OK.

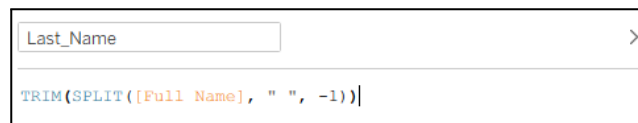


STEP 3 — Last_Name

1. Data Pane → Right Click
2. Create Calculated Field
3. Name:
Last_Name
4. Formula:

TRIM(SPLIT([Full_Name], " ", -1))

5. OK.



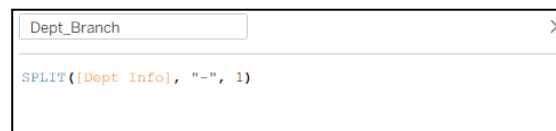
STEP 4 — Dept_Branch

Tumhare Dept_Info mein values jaise CSE-01, ECE-02 hain.

1. Create Calculated Field
2. Name:
Dept_Branch
3. Formula:

SPLIT([Dept_Info], "-", 1)

4. OK.



STEP 5 — Section_ID

1. Create Calculated Field
2. Name:
Section_ID
3. Formula:

SPLIT([Dept_Info], "-", 2)

4. OK.



STEP 6 — Student_Display_Label

1. Create Calculated Field
2. Name:
Student_Display_Label
3. Formula:

[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"

4. OK.

